

CHAPTER 12

NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS

12.1 INTRODUCTION

Differential equations arise frequently in many problems in mechanics, physics, chemistry, algebra, geometry, engineering, economics etc. Although many differential equations can be solved analytically but a great number of them cannot be so solved. However, the solution of many of these equations can usually be found numerically.

The general form of an n th order ordinary differential equation is

$$y^{(n)} = f(x, y, y', \dots, y^{(n-1)}). \quad (12.1)$$

It is called n th-order because the highest derivative is of order n , and ordinary because it involves only total derivatives and no partial derivatives (i.e. there is only one independent variable x). A function of the form $y = \phi(x)$ that satisfies (12.1) is called a solution of this equation. In general, there are many functions of the form $y = \phi(x)$, that satisfy (12.1). To get a unique solution, it is required to supply some additional information, e.g. values of $y(x)$ and (or) of its derivatives at some specific values of x . If all information is given at the same value of x , then the problem is called an initial-value problem. On the other hand, if the information is given for two or more values of x , the problem is called a boundary-value problem.

Various methods can be divided roughly into two groups, single-step and multistep methods. Single-step methods use information from only the last computed point (x_n, y_n) to give (x_{n+1}, y_{n+1}) . Multistep methods use information from several points (x_n, y_n) , (x_{n-1}, y_{n-1}) , ... for this purpose.

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DEFINITION 12.1: A function $f(x,y)$ is said to satisfy a Lipschitz condition on a set $D \subset \mathbb{R}^2$ in the variable y if there exists a constant $L > 0$ such that

$$|f(x,y_1) - f(x,y_2)| \leq L |y_1 - y_2|, \quad \forall (x,y_1), (x,y_2) \in D.$$

The constant L is called a Lipschitz constant for f . For example, if

$$f(x,y) = x|y| \quad \text{and} \quad D = \{(x,y) \mid 1 \leq x \leq 2, -3 \leq y \leq 4\},$$

then

$$\begin{aligned} |f(x,y_1) - f(x,y_2)| &= |x|y_1| - x|y_2||, \quad \text{for } (x,y_1), (x,y_2) \in D \\ &= |x||y_1| - |y_2|| \\ &= |x| \quad ||y_1| - |y_2|| \\ &\leq 2|y_1 - y_2|. \end{aligned}$$

Thus, f satisfies a Lipschitz condition in the variable y on D with Lipschitz constant 2.

The following theorem gives a sufficient condition for a solution of first-order ordinary differential equations. A proof of this theorem can be found in Birkhoff and Rota [2].

THEOREM 12.1: Let $f(x,y)$ be defined and continuous on a region D , where $D = \{(x,y) \mid a \leq x \leq b, -\infty < y < \infty\}$, a and b are finite. If f satisfies a Lipschitz condition on D in y ; then the initial-value problem

$$\frac{dy}{dx} = f(x,y), \quad y(a) = y_0,$$

has a unique solution $y(x)$ for $a \leq x \leq b$.

EXAMPLE 12.1: Discuss the existence of a unique solution of the initial-value problem

$$\frac{dy}{dx} = 1 + x \sin(xy), \quad 0 \leq x \leq 2, \quad y(0) = 0.$$

SOLUTION: Here $f(x,y) = 1 + x \sin(xy)$.

Keeping x constant and applying the mean value theorem (Appendix A) to the function $f(x,y)$, we find that whenever $y_1 < y_2$, there exists ξ , where $y_1 < \xi < y_2$ such that

$$\frac{\partial}{\partial y} f(x, \xi) = x^2 \cos(\xi x) = \frac{f(x, y_2) - f(x, y_1)}{y_2 - y_1}$$

or

$$|f(x, y_2) - f(x, y_1)| = |x^2 \cos(\xi x)| |y_2 - y_1| \leq 4 |y_2 - y_1|.$$

Thus, f satisfies a Lipschitz condition in the variable y with Lipschitz constant 4. Moreover, $f(x, y)$ is continuous for $0 < x < 2$ and $-\infty < y < \infty$. The Theorem 12.1 implies that the given initial value problem has a unique solution.

EXERCISE 12.1

1. Use Theorem 12.1 to show that

$$\frac{dy}{dx} = y \cos x, \quad 0 \leq x \leq 1, \quad y(0) = 1$$

has a unique solution. Find the solution.

2. Use Theorem 12.1 to show that

$$\frac{dy}{dx} = \frac{2}{x} y + x^2 e^x, \quad 1 \leq x \leq 2, \quad y(1) = 0$$

has a unique solution. Find the solution also.

12.2 EULER'S METHOD

The simplest method for the numerical solution of a first order ordinary differential equation is Euler's method. It is not very accurate and is seldom used in practice. A first-order ordinary differential equation may be written as

$$\frac{dy}{dx} = f(x, y). \quad (12.2)$$

The solution of (12.2) may be written symbolically as

$$y = \phi(x). \quad (12.3)$$

The graph of (12.3) in xy -plane is a curve; and since a smooth curve is practically straight for a small distance from any point

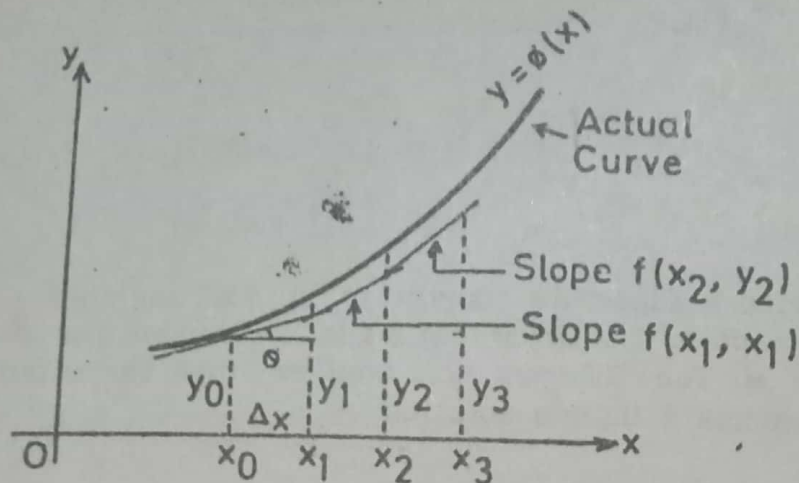


Fig. 12.1 Euler's method.

on it, we get from Fig. 12.1 the approximate relation

$$\begin{aligned}\Delta y &= \Delta x \tan \theta = \Delta x \left(\frac{dy}{dx} \right)_{(x_0, y_0)} \\ &= \Delta x f(x_0, y_0)\end{aligned}$$

$$\text{Thus } y_1 = y_0 + \Delta y \quad (12.4)$$

$$\text{or } y_1 = y_0 + \Delta x f(x_0, y_0).$$

If we use h in place of Δx then the above equation becomes

$$y_1 = y_0 + h f(x_0, y_0).$$

Taking $x_i = x_0 + ih$, $i = 0, 1, 2, \dots, n+1$

we get

$$y_2 = y_1 + h f(x_1, y_1)$$

$$y_3 = y_2 + h f(x_2, y_2)$$

and in general

$$y_{n+1} = y_n + h f(x_n, y_n). \quad (12.5)$$

This is known as Euler's method. The calculation work is

continued, interval by interval, along the x -axis from the starting point x_0 to the required finishing point. When Euler's method is applied repeatedly on several intervals in sequence, the numerical solution traces out a polygonal path with sides of slopes $f(x_n, y_n)$, $n = 0, 1, \dots, n-1$.

EXAMPLE 12.2: Find an approximate solution of the initial-value problem

$$y' = x - y, \quad 0 \leq x \leq 0.5, \quad y(0) = 1$$

taking $h = 0.1$.

SOLUTION: Using Euler's method we have

$$\begin{aligned} y_1 &= y_0 + hf(x_0, y_0), & f(x, y) &= x - y \\ &= 1 + 0.1(0 - 1) = 1 - 0.1 = 0.9 \end{aligned}$$

$$\begin{aligned} y_2 &= y_1 + hf(x_1, y_1) \\ &= 0.9 + 0.1(0.1 - 0.9) = 0.9 - 0.08 = 0.82 \end{aligned}$$

$$\begin{aligned} y_3 &= y_2 + hf(x_2, y_2) \\ &= 0.82 + 0.1(0.2 - 0.82) = 0.82 - 0.062 = 0.758 \end{aligned}$$

$$\begin{aligned} y_4 &= y_3 + hf(x_3, y_3) \\ &= 0.758 + 0.1(0.3 - 0.758) = 0.7122 \end{aligned}$$

$$\begin{aligned} y_5 &= y_4 + hf(x_4, y_4) \\ &= 0.7122 + 0.1(0.4 - 0.7122) = 0.68098. \end{aligned}$$

The exact solution of the given problem is $y = 2e^{-x} + x - 1$. The approximate values are compared with actual values and error in each case is also evaluated as shown in the following table:

Table 12.1 Results of Euler's Method

n	x_n	y_n App. value	$y(x_n)$ True value	$y(x_n) - y_n$ Error
0	0	1.0	1.0	0.0
1	0.1	0.9	0.9097	0.0097
2	0.2	0.82	0.8375	0.0175
3	0.3	0.758	0.7816	0.0236
4	0.4	0.7122	0.7406	0.0284
5	0.5	0.6810	0.7131	0.0321

EXERCISE 12.2

1. Use Euler's method with $h=0.1$ to approximate the solution of

a) $\frac{dy}{dx} = 1 + x \sin(xy), \quad 0 \leq x \leq 1, \quad y(0) = 0.$

b) $\frac{dy}{dx} = \frac{1}{2}x + y - 1, \quad 0 \leq x \leq 1, \quad y(0) = 1.$

2. Obtain an approximation to the solution $y(x) = e^{-x} + x$ to the initial-value problem

$$y' = -y + x + 1, \quad 0 \leq x \leq 1, \quad y(0) = 1,$$

using Euler's method with $h = 0.2$ and $h = 0.1$. Compare the approximations to the actual solution.

3. Compute an approximation to the solution $y(x) = -1/x$, using Euler's method with $h = 0.05$ to the initial-value problem

$$y' = \frac{1}{x^2} - \frac{y}{x} - y^2, \quad 1 \leq x \leq 1.5, \quad y(1) = -1.$$

Compare the approximation to the actual solution.

4. For the initial-value problem

$$\frac{dy}{dx} = 2x + y, \quad 0 \leq x \leq 1, \quad y(0) = 2,$$

approximate the solution to 4 decimal places, using Euler's method with $h = 0.2$.

5. Solve the differential equation

$$\frac{dy}{dx} = x^2 + y^2, \quad 0 \leq x \leq 1, \quad y(0) = 0$$

using Euler's method with $h = 0.1$ and working to 3 decimal places.

6. Approximate the initial-value problem

$$\frac{dy}{dx} = x + y, \quad 0 \leq x \leq 1, \quad y(0) = 1$$

to 4 decimal places with $h = 0.1$. Compare the results with the analytical solution $y = 2e^x - x - 1$ of the differential equation.

7. Solve the differential equation

$$\frac{dy}{dx} = e^x - e^{-x} - y \quad \text{with } y(0) = 1/2 \text{ for } 0 \leq x \leq 1.$$

- a) Use the Euler's method with $h = 0.1$.
- b) Use the Euler's method with $h = 0.2$.
- c) Given that the analytical solution is $y = (1/2)e^x - xe^{-x}$, find the error in each case.

8. Repeat problem 7 for the differential equation

$$\frac{dy}{dx} = 2y + x^2 e^x \quad \text{with } y(0) = -2$$

of which the analytical solution is $y = -(x^2 + 2x + 2)e^x$.

9. Obtain by Euler's method the numerical solution of

$$\frac{dy}{dx} = 2x - y, \quad 1 \leq x \leq 2, \quad y(1) = 3$$

with $h=0.1$. Compare your results with those obtained from the analytical solution $y = 2x + 3e^{1-x} - 2$.

12.3 IMPROVED EULER'S METHOD

Euler's Method is very slow when h is small and is inaccurate in case h is not small. The Euler's method gives Δy represented by CE (Fig. 12.2) which is an under-estimate of CB. If a line AD is drawn through A and parallel to the tangent at B, the Δy computed by Euler's method by using the slope at B is represented by CD which is an over-estimate of CB. It would therefore seem reasonable to take the average of these two slopes to give a better approximation to Δy than either taken alone. Using average slope, we get

$$\begin{aligned} \Delta y &= \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1)] \\ &= \frac{1}{2} [hf(x_0, y_0) + hf(x_1, y_1)] \\ &= \frac{1}{2} [CE + CD] = \frac{1}{2} [CE + CE + ED] \\ &= \frac{1}{2} [2CE + ED] = CE + \frac{1}{2} ED \end{aligned}$$

which is very close to its exact value CB. Using (12.4) we get

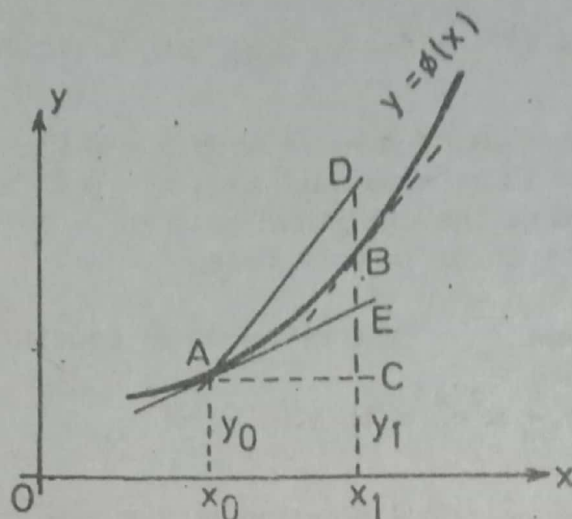


Fig.12.2 Improved Euler's method.

$$y_1 = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1)].$$

In general

$$y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1})]. \quad (12.6)$$

This is called improved Euler's method. It is an example of the procedure which is known as a predictor-corrector method. The improved Euler method (12.6) first requires us to find an initial estimate of y_{n+1} from (12.5). This is a relatively inaccurate formula compared with (12.6) and is used to predict an approximate value of y_{n+1} . This value of y_{n+1} is then used in (12.6) which gives a better approximation of y_{n+1} than the one obtained previously by using (12.5). In other words (12.6) is used to correct the initially predicted value of y_{n+1} . Due to this reason (12.5) is called predictor formula and (12.6) is called corrector formula.

Another modification of Euler's method makes use of the slope of the curve at midpoint of each interval to estimate Δy , i.e.

$$\begin{aligned} \Delta y &= h f\left(\frac{x_n + x_{n+1}}{2}, \frac{y_n + y_{n+1}}{2}\right) \\ &= h f\left(x_n + \frac{h}{2}, y_n + \frac{h}{2} f(x_n, y_n)\right) \end{aligned}$$

Thus

$$y_{n+1} = y_n + hf\left(x_n + \frac{h}{2}, y_n + \frac{h}{2} f(x_n, y_n)\right) \quad (12.7)$$

This is known as the midpoint method.

EXAMPLE 12.3: Solve the initial-value problem of example 12.2 by using improved Euler method.

SOLUTION: The computed values are tabulated below.

Table 12.2 Results of Improved Euler's Method Iteration

n	x_n	y_n	y_{n+1} Predicted value	y_{n+1} Corrected value	$y(x_{n+1})$ True value	$y(x_{n+1}) - y_{n+1}$ Error
0	0.0	1.0	0.9	0.91	0.9097	-0.0003
1	0.1	0.91	0.829	0.8380	0.8375	-0.0005
2	0.2	0.8380	0.7742	0.7824	0.7816	-0.0008
3	0.3	0.7824	0.7342	0.7416	0.7406	-0.0010
4	0.4	0.7416	0.7074	0.7142	0.7131	-0.0011
5	0.5	0.7142				

We note that errors are comparatively much smaller than those obtained previously.

12.4 MODIFIED EULER METHOD

The improved Euler's method can be improved further by an iterative process. The predicted value of y_{n+1} by (12.5) is used in (12.6) to improve the estimate (corrected value) for y_{n+1} . This corrected value of y_{n+1} is again used in (12.6) for further improvement. The process of correction is continued until no further change in the value of y_{n+1} is produced to the number of digits retained.

We now calculate the order of error involved in various Euler's methods. A method that agrees with Taylor's series of y_{n+1} upto and including the term in h^p is said to be pth order written as $O(h^p)$. By two-term Taylor's series

$$y_{n+1} = y_n + y'_n h + \frac{y''(\xi)}{2} h^2, \quad x_n < \xi < x_n + h$$

or
$$y_{n+1} = y_n + hy'_n + O(h^2), \quad \text{where } \frac{y''(\xi)}{2} h^2 = O(h^2).$$

Hence
$$y_{n+1} = y_n + hf(x_n, y_n) + O(h^2).$$

This shows that the error of each step of Euler's method is $O(h^2)$. This is called local error. The total error i.e. an accumulation of errors from step to step, is called global error. The global error of Euler's method is $O(h)$. Thus Euler's method is first order process.

Similarly, three-term Taylor's series is

$$y_{n+1} = y_n + hy'_n + \frac{h^2}{2} y''_n + \frac{y'''(\xi)}{6} h^3, \quad x_n < \xi < x_{n+1}$$

or
$$y_{n+1} = y_n + hy'_n + \frac{h^2}{2} y''_n + O(h^3)$$

Since
$$y''_n = \frac{y'_{n+1} - y'_n}{h} + O(h),$$

therefore
$$\begin{aligned} y_{n+1} &= y_n + hy'_n + \frac{1}{2} \left[\frac{y'_{n+1} - y'_n}{h} + O(h) \right] h^2 + O(h^3) \\ &= y_n + \frac{h}{2} [y'_n + y'_{n+1}] + O(h^3). \end{aligned}$$

Hence
$$y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1})] + O(h^3).$$

Thus the local error of the improved Euler method as well as modified Euler method is $O(h^3)$ and their truncation error is $O(h^2)$. These methods are called 2nd-order processes.

EXAMPLE 12.4: Solve example 12.2 by modified Euler method.

SOLUTION: We have already calculated $y_1 = 0.9$ by using (12.5) and $y_1 = 0.91$ by using (12.6). We write $y_1 = y_1^{(1)}$, the first approximation for y_1 . Substituting this value again in (12.6) we get 2nd approximation for y_1 .

Therefore,

$$\begin{aligned}
 y_1^{(2)} &= y_0 + \frac{0.1}{2} [f(x_0, y_0) + f(x_1, y_1)] \\
 &= 1 + 0.05[f(0, 1) + f(0.1, 0.91)] \\
 &= 1 + 0.05[0 - 1 + 0.1 - 0.91] = 0.9095.
 \end{aligned}$$

Similarly, the 3rd approximation for y_1 is

$$\begin{aligned}
 y_1^{(3)} &= y_0 + \frac{0.1}{2} [f(x_0, y_0) + f(x_1, y_1)] \\
 &= 1 + 0.05[f(0, 1) + f(0.1, 0.9095)] \\
 &= 1 + 0.05[0 - 1 + 0.1 - 0.9095] \\
 &= 0.909525 \approx 0.9095.
 \end{aligned}$$

Since the 3rd approximation for y_1 is same to 4 decimal places, the number of digits retained, therefore

$$y_1 = 0.9095.$$

Now

$$\begin{aligned}
 y_2 &= y_1 + hf(x_1, y_1) = 0.9095 + 0.1(0.1 - 0.9095) \\
 &= 0.82855 \approx 0.8286.
 \end{aligned}$$

and

$$\begin{aligned}
 y_2^{(1)} &= y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2)] \\
 &= 0.9095 + 0.05[f(0.1, 0.9095) + f(0.2, 0.8286)] \\
 &= 0.9095 + 0.05[0.1 - 0.9095 + 0.2 - 0.8286] \\
 &= 0.837595 \approx 0.8376.
 \end{aligned}$$

$$\begin{aligned}
 y_2^{(2)} &= 0.9095 + 0.05[f(0.1, 0.9095) + f(0.2, 0.8376)] \\
 &= 0.9095 + 0.05[0.1 - 0.9095 + 0.2 - 0.8376] \\
 &= 0.837145 \approx 0.8371.
 \end{aligned}$$

$$\begin{aligned}
 y_2^{(3)} &= 0.9095 + 0.05[0.1 - 0.9095 + 0.2 - 0.8371] \\
 &= 0.83717 \approx 0.8372.
 \end{aligned}$$

$$\begin{aligned}
 y_2^{(4)} &= 0.9095 + 0.05[0.1 - 0.9095 + 0.2 - 0.8372] \\
 &= 0.837165 \approx 0.8372.
 \end{aligned}$$

Therefore

$$y_2 = 0.8372.$$

Similarly

$$y_3 = 0.8372 + 0.1[0.2 - 0.8372] = 0.77348 \approx 0.7735.$$

and

$$\begin{aligned}
 y_3^{(1)} &= 0.8372 + 0.05[(0.2 - 0.8372) + (0.3 - 0.7735)] \\
 &= 0.781665 \approx 0.7817.
 \end{aligned}$$

$$y_3^{(2)} = 0.8372 + 0.05[(0.2 - 0.8372) + (0.3 - 0.7817)] \\ = 0.781255 \approx 0.7813.$$

$$y_3^{(3)} = 0.8372 + 0.05[(0.2 - 0.8372) + (0.3 - 0.7813)] \\ = 0.781275 \approx 0.7813.$$

Hence

$$y_3 = 0.7813.$$

Continuing in this way we get recorrected values

$$y_4 = 0.7402 \text{ and } y_5 = 0.7126.$$

In this case, the errors in the values of y from $x = 0$ to $x = 0.5$ respectively are, 0, 0.0002, 0.0003, 0.0003, 0.0004, 0.0005. These errors in magnitude are approximately one half of those introduced by the improved Euler method.

EXERCISE 12.3

1. Solve the differential equation

$$\frac{dy}{dx} = (x+y)^{1/2}, \quad 0 \leq x \leq 0.3, \quad y(0) = 0.3600$$

with $h = 0.1$ by

- a) Improved Euler's method.
- b) Modified Euler's method.

2. Solve the differential equation

$$\frac{dy}{dx} = x + y, \quad 0 \leq x \leq 0.2, \quad y(0) = 1$$

taking $h = 0.1$ and working to 4 decimal places by

- a) Improved Euler's method.
- b) Modified Euler's method.

Compare the results with the analytical solution $y = 2e^x - x - 1$ of the differential equation.

3. Solve

$$\frac{dy}{dx} = 2 \cos x + y, \quad 0 \leq x \leq 0.2, \quad y(0) = -1$$

with $h = 0.1$ and using

- a) Improved Euler's method.
- b) Modified Euler's method.

4. Solve the differential equation

$$\frac{dy}{dx} = 2y + x^2 e^x; 0 \leq x \leq 0.3, y(0) = -2,$$

using

- a) Improved Euler's method,
- b) Modified Euler's method,

with $h = 0.1$.

12.5 RUNGE-KUTTA METHODS

Runge-Kutta methods are also based on (12.4) but use a weighted average of several estimates of Δy to give a more accurate approximation. The second order Runge-Kutta method will be developed in detail. The derivation of the higher order is complicated but can be derived in a similar way.

Suppose we are solving an equation of the form

$$\frac{dy}{dx} = f(x, y).$$

Let Δy be a weighted average of two derivative evaluations k_1 , and k_2 on the interval $x_n \leq x \leq x_{n+1}$, that is,

$$\Delta y = ak_1 + bk_2$$

where

$$\begin{aligned} k_1 &= hf(x_n, y_n) \\ k_2 &= hf(x_n + rh, y_n + sk_1) \end{aligned} \quad (12.8)$$

and then

$$y_{n+1} = y_n + \Delta y = y_n + ak_1 + bk_2 \quad (12.9)$$

where a, b, r, s are constants chosen so that (12.9) agrees as closely as possible with Taylor's series

$$y_{n+1} = y(x_n + h) = y_n + hy'_n + \frac{h^2}{2!} y''_n + \frac{h^3}{3!} y'''_n + \dots$$

$$\text{or } y_{n+1} = y_n + hf(x_n, y_n) + \frac{h^2}{2!} f'(x_n, y_n) + \frac{h^3}{3!} f''(x_n, y_n) + \dots \quad (12.10)$$

where we have used the differential equation

$$y' = f(x, y) \text{ or } y'_n = f(x_n, y_n)$$

and so
$$y''_n = \frac{df}{dx}(x_n, y_n) = f'_x(x_n, y_n), \quad y'''_n = f''(x_n, y_n).$$

Using the chain rule

$$\begin{aligned} \frac{df}{dx} &= f'_x(x_n, y_n) + f'_y(x_n, y_n) \frac{dy}{dx} \\ &= f'_x(x_n, y_n) + f'_y(x_n, y_n) f(x_n, y_n). \end{aligned}$$

Similarly,

$$\begin{aligned} y'''_n &= \frac{d}{dx} [f'_x(x_n, y_n) + f'_y(x_n, y_n) f(x_n, y_n)] \\ y'''_n &= f''_{xx}(x_n, y_n) + 2f''_{xy}(x_n, y_n) f(x_n, y_n) + f''_{yy}(x_n, y_n) f^2(x_n, y_n) \\ &\quad + f'_y(x_n, y_n) f^2(x_n, y_n) + f'^2_y(x_n, y_n) f(x_n, y_n). \end{aligned}$$

Substituting these values in (12.10), we obtain

$$y_{n+1} = y_n + hf(x_n, y_n) + \frac{h^2}{2} [f''_{xx}(x_n, y_n) + 2f''_{xy}(x_n, y_n) f(x_n, y_n) + f''_{yy}(x_n, y_n) f^2(x_n, y_n) + \dots] \quad (12.11)$$

Expanding k_2 in a Taylor's series for a function of two variables (see Appendix A) gives

$$\begin{aligned} k_2 &= h[f'_x(x_n, y_n) + rh f''_{xx}(x_n, y_n) + sk_1 f''_{xy}(x_n, y_n) + \dots] \\ &= h[f'_x(x_n, y_n) + h(r f''_{xx}(x_n, y_n) + s f''_{xy}(x_n, y_n) f(x_n, y_n) + \dots)] \end{aligned}$$

Substituting the values of k_1 and k_2 in (12.9), we get

$$\begin{aligned} y_{n+1} &= y_n + ahf(x_n, y_n) + bh[f'_x(x_n, y_n) + h(r f''_{xx}(x_n, y_n) + s f''_{xy}(x_n, y_n) f(x_n, y_n) + \dots)] \\ &= y_n + h(a+b)f(x_n, y_n) + h^2[br f''_{xx}(x_n, y_n) + sb f''_{xy}(x_n, y_n) f(x_n, y_n) + \dots] \quad (12.12) \end{aligned}$$

Comparing (12.11) and (12.12) we obtain

$$\begin{aligned} a + b &= 1, \\ br &= 1/2, \\ bs &= 1/2. \end{aligned}$$

because the four unknowns have to satisfy only three equations, the value of one of them can be chosen arbitrarily provided the resulting equations have a solution. This shows that there are infinite number of second-order Runge-Kutta methods each of which is $O(h^2)$. The two common choices are $b = \frac{1}{2}$ and $b = 1$.

For $b = \frac{1}{2}$, $a = \frac{1}{2}$, $r = s = 1$. Then (12.9) becomes

$$y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1})]$$

which is the improved Euler method.

For $b = 1$, $a = 0$, $r = s = 1/2$. Then (12.9) becomes

$$y_{n+1} = y_n + hf\left(x_n + \frac{h}{2}, y_n + \frac{h}{2} f(x_n, y_n)\right)$$

which is the midpoint method.

If we choose $b = 3/4$ then $a = 1/4$, $r=s=2/3$. In this case (12.9) becomes

$$y_{n+1} = y_n + \frac{1}{4} (k_1 + 3k_2)$$

$$y_{n+1} = y_n + \frac{h}{4} \left(f(x_n, y_n) + 3f\left(x_n + \frac{2}{3}h, y_n + \frac{2}{3}hf(x_n, y_n)\right) \right).$$

This is known as the Heun's method.

When Heun's method is applied to an initial-value problem

$$y' = x^2 + y^2, y(0) = 0 \text{ with } h = 0.1,$$

it calculates $y(1) \approx 0.349640$, whereas the true value is $y(1) \approx 0.350232$. Euler's method calculates $y(1) \approx 0.292542$. However, Euler's method requires only 10 times calculation of the function $f(x, y)$, whereas Heun's method requires 20 times. At $h = 0.05$, Euler's method also requires 20 times computation of $f(x, y)$ and gives the approximation $y(1) \approx 0.3202117$, which still has about 50 times the error of the Heun's method.

The higher-order Runge-Kutta methods are derived in similar way. In this case

$$\Delta y = ak_1 + bk_2 + ck_3$$

where

$$k_1 = hf(x_n, y_n)$$

Because the four unknowns have to satisfy only three equations, the value of one of them can be chosen arbitrarily provided the resulting equations have a solution. This shows that there are infinite number of second-order Runge-Kutta methods each of which is $O(h^2)$. The two common choices are $b = \frac{1}{2}$ and $b = 1$.

For $b = \frac{1}{2}$, $a = \frac{1}{2}$, $r = s = 1$. Then (12.9) becomes

$$y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1})]$$

which is the improved Euler method.

For $b = 1$, $a = 0$, $r = s = 1/2$. Then (12.9) becomes

$$y_{n+1} = y_n + hf\left(x_n + \frac{h}{2}, y_n + \frac{h}{2} f(x_n, y_n)\right)$$

which is the midpoint method.

If we choose $b = 3/4$ then $a = 1/4$, $r=s=2/3$. In this case (12.9) becomes

$$y_{n+1} = y_n + \frac{1}{4} (k_1 + 3k_2)$$

or
$$y_{n+1} = y_n + \frac{h}{4} \left(f(x_n, y_n) + 3f\left(x_n + \frac{2}{3}h, y_n + \frac{2}{3}hf(x_n, y_n)\right) \right).$$

This is known as the Heun's method.

When Heun's method is applied to an initial-value problem

$$y' = x^2 + y^2, y(0) = 0 \text{ with } h = 0.1,$$

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The higher-order Runge-Kutta methods are derived in similar way. In this case

$$\Delta y = ak_1 + bk_2 + ck_3$$

where

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf(x_n + qh, y_n + qk_1)$$

$$k_3 = hf(x_n + rh, y_n + sk_2 + (r - s)k_1).$$

Then third-order Runge-Kutta methods are given by

$$y_{n+1} = y_n + ak_1 + bk_2 + ck_3. \quad (12.13)$$

We first expand k_2 and k_3 about (x_n, y_n) in Taylor's series as a function of two variables. Coefficients of like powers of h through the h^3 terms in (12.10) and (12.13) are compared to get a formula with a local error $O(h^4)$. The four equations in six unknowns obtained in this way are

$$\begin{aligned} a + b + c &= 1 \\ bq + cr &= 1/2 \\ bq^2 + cr^2 &= 1/3 \\ cqs &= 1/6. \end{aligned}$$

Two out of six constants a, b, c, q, r and s are arbitrary. The third-order Runge-Kutta method with constants selected by Kutta is

$$\left. \begin{aligned} y_{n+1} &= y_n + \frac{h}{6} (k_1 + 4k_2 + k_3) \\ k_1 &= f(x_n, y_n) \\ k_2 &= f(x_n + \frac{h}{2}, y_n + \frac{h}{2} k_1) \\ k_3 &= f(x_n + h, y_n + 2hk_2 - hk_1) \end{aligned} \right\} \quad (12.14)$$

If $\frac{dy}{dx} = f(x)$, i.e., dy/dx is a function of x only, then Runge-Kutta method reduces to Simpson's $1/3$ rule, Eq.(11.9). In this case

$$\begin{aligned} k_1 &= f(x_n), \\ k_2 &= f(x_n + \frac{h}{2}), \\ k_3 &= f(x_n + h), \end{aligned}$$

and therefore

$$\Delta y = \frac{h}{6} \left[f(x_n) + 4f(x_n + \frac{h}{2}) + f(x_n + h) \right]$$

$$\Delta y = \frac{h/2}{3} \left[f(x_n) + 4f\left(x_n + \frac{h}{2}\right) + f(x_n + h) \right]$$

which is the same result as would be obtained by applying Simpson's 1/3 rule on the interval $[x_n, x_n + h]$ if we take two equal subintervals of width $h/2$.

The fourth-order Runge-Kutta methods are most widely used in computer solutions to differential equations. Comparing coefficients of like power of h through the h^4 terms gives a set of eleven equations in thirteen unknowns. The eleven equations can be solved with two unknowns chosen arbitrarily. Several fourth-order algorithms are used. The most commonly used fourth-order method is

$$\left. \begin{aligned} y_{n+1} &= y_n + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) \\ k_1 &= hf(x_n, y_n) \\ k_2 &= hf\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right) \\ k_3 &= hf\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right) \\ k_4 &= hf(x_n + h, y_n + k_3) \end{aligned} \right\} \quad (12.15)$$

The local error of the fourth-order Runge-Kutta method is $O(h^5)$. We note that (12.15) again reduces to Simpson's 1/3 rule if $f(x, y)$ is a function of x only.

EXAMPLE 12.5: Solve the initial-value problem of example 12.2 using fourth-order Runge-Kutta method.

SOLUTION: We have

$$\begin{aligned} k_1 &= hf(x_0, y_0) = 0.1 f(0, 1) = 0.1 (0 - 1) = -0.1 \\ k_2 &= hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.1 f(0.05, 1 - 0.05) \\ &= 0.1 f(0.05, 0.95) = 0.1 (0.05 - 0.95) = -0.09 \\ k_3 &= hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = 0.1 f(0.05, 1 - 0.045) \\ &= 0.1 f(0.05, 0.955) = 0.1 (0.05 - 0.955) = -0.0905 \\ k_4 &= hf(x_0 + h, y_0 + k_3) = 0.1 f(0.1, 0.9095) \\ &= 0.1 (0.1 - 0.9095) = 0.1 (-0.8095) = -0.08095 \end{aligned}$$

$$k_2 = hf(x_n + qh, y_n + qk_1)$$

$$k_3 = hf(x_n + rh, y_n + sk_2 + (r - s)k_1).$$

Then third-order Runge-Kutta methods are given by

$$y_{n+1} = y_n + ak_1 + bk_2 + ck_3. \quad (12.13)$$

We first expand k_2 and k_3 about (x_n, y_n) in Taylor's series as a function of two variables. Coefficients of like powers of h through the h^3 terms in (12.10) and (12.13) are compared to get a formula with a local error $O(h^4)$. The four equations in six unknowns obtained in this way are

$$a + b + c = 1$$

$$bq + cr = 1/2$$

$$bq^2 + cr^2 = 1/3$$

$$cqs = 1/6.$$

Two out of six constants a, b, c, q, r and s are arbitrary. The third-order Runge-Kutta method with constants selected by Kutta is

$$\left. \begin{aligned} y_{n+1} &= y_n + \frac{h}{6} (k_1 + 4k_2 + k_3) \\ k_1 &= f(x_n, y_n) \\ k_2 &= f(x_n + \frac{h}{2}, y_n + \frac{h}{2} k_1) \\ k_3 &= f(x_n + h, y_n + 2hk_2 - hk_1) \end{aligned} \right\} \quad (12.14)$$

If $\frac{dy}{dx} = f(x)$, i.e., dy/dx is a function of x only, then Runge-Kutta method reduces to Simpson's $1/3$ rule, Eq.(11.9). In this case

$$k_1 = f(x_n),$$

$$k_2 = f(x_n + \frac{h}{2}),$$

$$k_3 = f(x_n + h),$$

and therefore

$$\Delta y = \frac{h}{6} \left[f(x_n) + 4f(x_n + \frac{h}{2}) + f(x_n + h) \right]$$

$$\Delta y = \frac{h/2}{3} \left[f(x_n) + 4f\left(x_n + \frac{h}{2}\right) + f(x_n + h) \right]$$

which is the same result as would be obtained by applying Simpson's 1/3 rule on the interval $[x_n, x_n + h]$ if we take two equal subintervals of width $h/2$.

The fourth-order Runge-Kutta methods are most widely used in computer solutions to differential equations. Comparing coefficients of like power of h through the h^4 terms gives a set of eleven equations in thirteen unknowns. The eleven equations can be solved with two unknowns chosen arbitrarily. Several fourth-order algorithms are used. The most commonly used fourth-order method is

$$\left. \begin{aligned} y_{n+1} &= y_n + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) \\ k_1 &= hf(x_n, y_n) \\ k_2 &= hf\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right) \\ k_3 &= hf\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right) \\ k_4 &= hf(x_n + h, y_n + k_3) \end{aligned} \right\} \quad (12.15)$$

The local error of the fourth-order Runge-Kutta method is $O(h^5)$. We note that (12.15) again reduces to Simpson's 1/3 rule if $f(x, y)$ is a function of x only.

EXAMPLE 12.5: Solve the initial-value problem of example 12.2 using fourth-order Runge-Kutta method.

SOLUTION: We have

$$\begin{aligned} k_1 &= hf(x_0, y_0) = 0.1 f(0, 1) = 0.1 (0 - 1) = -0.1 \\ k_2 &= hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.1 f(0.05, 1 - 0.05) \\ &= 0.1 f(0.05, 0.95) = 0.1 (0.05 - 0.95) = -0.09 \\ k_3 &= hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = 0.1 f(0.05, 1 - 0.045) \\ &= 0.1 f(0.05, 0.955) = 0.1 (0.05 - 0.955) = -0.0905 \\ k_4 &= hf(x_0 + h, y_0 + k_3) = 0.1 f(0.1, 0.9095) \\ &= 0.1 (0.1 - 0.9095) = 0.1 (-0.8095) = -0.08095 \end{aligned}$$

Therefore
$$y_1 = y_0 + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

$$= 1 + \frac{1}{6} [-0.1 - 0.18 - 0.1810 - 0.08095] = 0.90968.$$

Similarly, for the next step

$$k_1 = hf(x_1, y_1) = 0.1 f(0.1, 0.90968) = 0.1(0.1 - 0.90968) = -0.08097.$$

$$k_2 = hf(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}) = 0.1 f(0.15, 0.85920) = -0.07192$$

$$k_3 = hf(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}) = 0.1 f(0.15, 0.87372) = -0.07237$$

$$k_4 = hf(x_1 + h, y_1 + k_3) = 0.1 f(0.2, 0.83731) = -0.06373$$

$$y_2 = y_1 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$= 0.90968 + \frac{1}{6} [-0.08097 + 2(-0.07192) + 2(-0.07237) - 0.06373]$$

$$= 0.83747.$$

Now for the third step we find

$$k_1 = hf(x_2, y_2) = 0.1 f(0.2, 0.83747) = -0.06375$$

$$k_2 = hf(x_2 + \frac{h}{2}, y_2 + \frac{k_1}{2}) = 0.1 f(0.25, 0.80560) = -0.05556$$

$$k_3 = hf(x_2 + \frac{h}{2}, y_2 + \frac{k_2}{2}) = 0.1 f(0.25, 0.80969) = -0.05597$$

$$k_4 = hf(x_2 + h, y_2 + k_3) = 0.1 f(0.3, 0.7815) = -0.04815$$

$$y_3 = y_2 + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4] = 0.78164.$$

Continuing this process, we get $y_4 = 0.74064$, $y_5 = 0.71306$.

We note that these values agree with the true values.

EXERCISE 12.4

1. Solve the initial-value problem

$$\frac{dy}{dx} = \frac{1}{2}(x + y) - 1, \quad 0 \leq x \leq 0.3, \quad y(0) = 1$$

using fourth-order Runge-Kutta method with $h = 0.1$. Compare your results with the analytical solution $y = e^{x/2} - x$ to this differential equation.

2. Use the fourth-order Runge-Kutta method with $h = 0.1$ to approximate the initial-value problem

$$\frac{dy}{dx} = 1 - x^2 + y^2, \quad 1 \leq x \leq 1.4, \quad y(1) = 0.$$

3. Use the fourth-order Runge-Kutta method with $h = 0.1$ to approximate the solution to

$$\frac{dy}{dx} = \frac{2y}{x} + x^2 e^x, \quad 1 \leq x \leq 1.4, \quad y(1) = 0.$$

4. Use the fourth-order Runge-Kutta method to find the solution to the following initial-value problems:

a) $\frac{dy}{dx} = -2y + 2x^2 + 2x, \quad 0 \leq x \leq 0.4, \quad y(0) = 1, \quad h = 0.2.$

b) $\frac{dy}{dx} = x - y + 1, \quad 0 \leq x \leq 0.5, \quad y(0) = 1, \quad h = 0.1.$

c) $\frac{dy}{dx} = 1 - y, \quad 0 \leq x \leq 0.3, \quad y(0) = 0, \quad h = 0.1.$

5. Apply the fourth-order Runge-Kutta method to find an approximate solution to the following initial-value problems. Compare your computed results with the analytical results:

a) $\frac{dy}{dx} + y - e^{-x} \cos x = 0, \quad 0 \leq x \leq 0.2, \quad y(0) = 0, \quad h = 0.1.$

Analytical solution is $y = e^{-x} \sin x$.

b) $y' = x^2 + y, \quad 0 \leq x \leq 0.2, \quad y(0) = 1, \quad h = 0.1.$

Analytical solution is $y = 3e^x - (x^2 + 2x + 2)$.

c) $y' = x^2 y, \quad 0 \leq x \leq 0.3, \quad y(0) = 1, \quad h = 0.1.$

Analytical solution is $y = e^{x^3/3}$.

12.6 PREDICTOR-CORRECTOR (P-C) METHODS

The Improved Euler method is an example of a more general class of methods known as predictor-corrector methods. All the methods discussed so far are the single-step methods while the predictor-corrector methods are multistep methods, i.e. they use information from several points (x_n, y_n) , (x_{n-1}, y_{n-1}) , ... to evaluate (x_{n+1}, y_{n+1}) . There are many ways of constructing these methods. In one of these, we integrate the differential equation (12.2) between the limits x_{n-k} and x_{n+1} (where k is any integer) and this gives

$$y_{n+1} = y_{n-k} + \int_{x_{n-k}}^{x_{n+1}} f(x, y) dx. \quad (12.16)$$

Substituting $x = x_n + ph$, (12.16) becomes

$$y_{n+1} = y_{n-k} + h \int_{-k}^1 f_p dp \quad (12.17)$$

where

$$f_p = f(p, y(p)) = f(x_n + hp, y(x_n + hp)).$$

Since

$$\begin{aligned} f_p &= E^p f_n \\ &= (1 - \nabla)^{-p} f_n \\ &= \left\{ 1 + p\nabla + \frac{1}{2}p(p+1)\nabla^2 + \frac{1}{3!}p(p+1)(p+2)\nabla^3 + \dots + \frac{1}{r!}p(p+1)\dots(p+r-1)\nabla^r + \dots \right\} f_n \end{aligned}$$

Therefore

$$\begin{aligned} y_{n+1} &= y_{n-k} + h \int_{-k}^1 \left\{ 1 + p\nabla + \frac{1}{2}p(p+1)\nabla^2 + \dots + \frac{1}{r!}p(p+1)\dots(p+r-1)\nabla^r + \dots \right\} f_n dp \\ \text{or} \quad y_{n+1} &= y_{n-k} + h \left[\left\{ p + \frac{p^2}{2}\nabla + \frac{1}{2}\left(\frac{p^3}{3} + \frac{p^2}{2}\right)\nabla^2 + \frac{1}{3!}\left(\frac{p^4}{4} + p^3 + p^2\right)\nabla^3 + \dots \right\} f_n \right]_{-k}^1 \\ y_{n+1} &= y_{n-k} + h \left\{ (1+k) + \frac{1}{2}(1-k^2)\nabla + \frac{1}{12}(5 - 3k^2 + 2k^3)\nabla^2 + \right. \\ &\quad \left. \frac{1}{24}(9 - 4k^2 + 4k^3 - k^4)\nabla^3 + \dots \right\} f_n. \end{aligned} \quad (12.18)$$

A variety of integration formulae can be obtained by choosing a value of k and truncating the series (12.18) after a selected number of terms. These allow y_{n+1} to be computed in terms of h , y_{n-k} , f_n and backward differences of f_n . For example, the choice of $k = 0$ and the truncation of the series after first term inside the parentheses of (12.18) leads to

$$y_{n+1} = y_n + h f_n$$

which is the Euler's method.

A corrector formula involves y_{n+1} on the right hand side as well as the left hand side. To obtain such a formula, we express f_p in terms of backward difference formula based at x_{n+1} instead of x_n

$$y_{n+1} = y_{n-k} + h \int_{-k}^1 \left[1 + (p-1)\nabla + \frac{1}{2!}(p-1)p\nabla^2 + \dots + \frac{1}{r!}(p-1)p(p+1)\dots (p+r-2)\nabla^r \dots \right] f_{n+1} dp$$

or

$$y_{n+1} = y_{n-k} + h \left[\left\{ p + \left(\frac{p^2}{2} - p \right) \nabla + \frac{1}{2!} \left(\frac{p^3}{3} - \frac{p^2}{2} \right) \nabla^2 + \frac{1}{3!} \left(\frac{p^4}{4} - \frac{p^3}{2} \right) \nabla^3 + \dots \right\} f_{n+1} \right]_{-k}^1$$

$$y_{n+1} = y_{n-k} + h(k+1) \left\{ 1 - \frac{(k+1)}{2} \nabla + \frac{1}{12} (2k^2 + k - 1) \nabla^2 - \frac{1}{24} (k+1)(k-1)^2 \nabla^3 \dots \right\} f_{n+1}$$

(12.19)

As before, a variety of integration formulae can be obtained by choosing a value for k and truncating the series (12.19) after a selected number of terms. In this case y_{n+1} is given in terms of h , y_{n-k} , f_{n+1} and backward differences of f_{n+1} . A corrector formula of any desired order of accuracy can be obtained by using (12.19). For example, the choice of $k = 0$ and truncation of the series after two terms inside the parentheses of (12.19) lead to

$$y_{n+1} = y_n + h(1 - \nabla/2)f_{n+1}$$

or

$$y_{n+1} = y_n + (h/2)(f_n + f_{n+1})$$

which is the corrector formula of the Euler's method.

A variety of integration formulae can be obtained by choosing a value of k and truncating the series (12.18) after a selected number of terms. These allow y_{n+1} to be computed in terms of h , y_{n-k} , f_n and backward differences of f_n . For example, the choice of $k = 0$ and the truncation of the series after first term inside the parentheses of (12.18) leads to

$$y_{n+1} = y_n + h f_n$$

which is the Euler's method.

A corrector formula involves y_{n+1} on the right hand side as well as the left hand side. To obtain such a formula, we express f_p in terms of backward difference formula based at x_{n+1} instead of x_n

$$y_{n+1} = y_{n-k} + h \int_{-k}^1 \left[1 + (p-1)\nabla + \frac{1}{2!}(p-1)p\nabla^2 + \dots + \frac{1}{r!}(p-1)p(p+1)\dots(p+r-2)\nabla^r \dots \right] f_{n+1} dp$$

or

$$y_{n+1} = y_{n-k} + h \left[\left\{ p + \left(\frac{p^2}{2} - p \right) \nabla + \frac{1}{2!} \left(\frac{p^3}{3} - \frac{p^2}{2} \right) \nabla^2 + \frac{1}{3!} \left(\frac{p^4}{4} - \frac{p^2}{2} \right) \nabla^3 + \dots \right\} f_{n+1} \right]_{-k}^1$$

$$y_{n+1} = y_{n-k} + h(k+1) \left\{ 1 - \frac{(k+1)}{2} \nabla + \frac{1}{12} (2k^2 + k - 1) \nabla^2 - \frac{1}{24} (k+1)(k-1)^2 \nabla^3 \dots \right\} f_{n+1}$$

(12.19)

As before, a variety of integration formulae can be obtained by choosing a value for k and truncating the series (12.19) after a selected number of terms. In this case y_{n+1} is given in terms of h , y_{n-k} , f_{n+1} and backward differences of f_{n+1} . A corrector formula of any desired order of accuracy can be obtained by using (12.19). For example, the choice of $k = 0$ and truncation of the series after two terms inside the parentheses of (12.19) lead to

$$y_{n+1} = y_n + h(1 - \nabla/2)f_{n+1}$$

or

$$y_{n+1} = y_n + (h/2)(f_n + f_{n+1})$$

which is the corrector formula of the Euler's method.

In practice we select two formulae with comparable order for local truncation error. One of these formulae is selected from (12.18) and the other is selected from (12.19). The formula selected from (12.18) is used to predict the value y_{n+1} and is called the predictor. The formula selected from (12.19) is then used to correct the value y_{n+1} and is called the corrector.

12.6.1 MILNE'S METHOD

A predictor-corrector method, known as Milne's method, is obtained from (12.18) and (12.19) by retaining three terms inside parentheses and setting $k=3$ in (12.18) whereas $k=1$ in (12.19). The resulting formulae are:

Predictor:

$$y_{n+1} = y_{n-3} + h(4 - 4\nabla + \frac{8}{3} \nabla^2) f_n \quad (12.20)$$

or

$$y_{n+1} = y_{n-3} + \frac{4}{3} h(2f_n - f_{n-1} + 2f_{n-2}). \quad (12.20')$$

corrector:

$$y_{n+1} = y_{n-1} + 2h(1 - \nabla + \frac{1}{6} \nabla^2) f_{n+1} \quad (12.21)$$

or

$$y_{n+1} = y_{n-1} + \frac{h}{3} (f_{n+1} + 4f_n + f_{n-1}). \quad (12.21')$$

The truncation errors of (12.20) and (12.21) are the first neglected terms in the series. But due to particular values chosen for k , the next terms are zero. In fact, the truncation errors of each formula are proportional to $h\nabla^4 f$. Using $\nabla \approx hD$, this is equivalent to $h^5 D^4 f$. Thus the local truncation error of each formula is of order h^5 whereas the global errors are of order h^4 .

12.6.2 ADAMS-MOULTON METHOD

Another predictor-corrector method, known as Adams-Moulton method, is obtained from (12.18) and (12.19) by retaining four terms from each of these formulae and setting $k=0$. The resulting formulae are:

Predictor:

$$y_{n+1} = y_n + h \left(f_n + \frac{1}{2} \nabla f_n + \frac{5}{12} \nabla^2 f_n + \frac{9}{24} \nabla^3 f_n \right) \quad (12.22)$$

or

$$y_{n+1} = y_n + \frac{h}{24} (55f_n - 59f_{n-1} + 37f_{n-2} - 9f_{n-3}). \quad (12.22')$$

Corrector:

$$y_{n+1} = y_n + h \left(f_{n+1} - \frac{1}{2} \nabla f_{n+1} + \frac{1}{12} \nabla^2 f_{n+1} - \frac{1}{24} \nabla^3 f_{n+1} \right) \quad (12.23)$$

or

$$y_{n+1} = y_n + \frac{h}{24} (9f_{n+1} + 19f_n - 5f_{n-1} + f_{n-2}). \quad (12.23')$$

As in Milne's method, the local truncation errors of these formulae are of order h^5 and the global errors are of order h^4 . Like the Runge-Kutta method, it is one of the reliable and most widely used procedures for the numerical solution of ordinary differential equations.

Predictor-corrector methods have one drawback that they are not self-starting. For example, if we use (12.20'), then we require the values of y at x_{n-3} , x_{n-2} , x_{n-1} and x_n . These values, speaking generally, are not provided in the initial conditions of the problem. Starting values, therefore, must be obtained by some other method. The modified Euler's method which is a single-step method, for example, can be used to obtain (x_1, y_1) , (x_2, y_2) , (x_3, y_3) . However, for a given mesh size it is relatively inaccurate and, as a consequence, the further values calculated using any higher order predictor-corrector method would be based on poor quality starting data having an inherited error. To overcome this a smaller step size must be used while finding the starting values.

EXAMPLE 12.6: Solve the initial-value problem of example 12.2 using

- a) Milne's method.
- b) Adams-Moulton method.

SOLUTION: We have already computed the values y_1 , y_2 , y_3 and so these values can be taken as starting values of these methods. The best values computed numerically so far are due to the fourth-order Runge-Kutta method. These values are:

$$y_0 = 1, y_1 = 0.90968, y_2 = 0.83747 \text{ and } y_3 = 0.78164.$$

a) Milne's method:

$$\begin{aligned} y_4^p &= y_0 + \frac{4}{3} h \left[2f_3 - f_2 + 2f_1 \right] \\ &= 1 + \frac{0.4}{3} \left[2f(x_3, y_3) - f(x_2, y_2) + 2f(x_1, y_1) \right] \\ &= 1 + \frac{0.4}{3} \left[2(0.3 - 0.78164) - (0.2 - 0.83747) + 2(0.1 - 0.90968) \right] \\ &= 1 - 0.25936 = 0.74075. \end{aligned}$$

$$\begin{aligned} y_4^c &= y_2 + \frac{h}{3} \left[f_4 + 4f_3 + f_2 \right] \\ &= y_2 + \frac{h}{3} \left[f(x_4, y_4^p) + 4f(x_3, y_3) + f(x_2, y_2) \right] \\ &= 0.83747 + \frac{0.1}{3} \left[(0.4 - 0.74075) + 4(0.3 - 0.78164) + (0.2 - 0.83747) \right] \\ &= 0.74064. \end{aligned}$$

Similarly,

$$\begin{aligned} y_5^p &= y_1 + \frac{4}{3} h \left[2f_4 - f_3 + 2f_2 \right] \\ &= 0.90968 + \frac{0.4}{3} \left[2(0.4 - 0.74064) - (0.3 - 0.78164) + 2(0.2 - 0.83747) \right] \\ &= 0.90968 - 0.19661 = 0.71307. \end{aligned}$$

$$\begin{aligned} y_5^c &= y_3 + \frac{h}{3} \left[f_5 + 4f_4 + f_3 \right] \\ &= 0.78164 + \frac{0.1}{3} \left[(0.5 - 0.71307) + 4(0.4 - 0.74064) + (0.3 - 0.78164) \right] \\ &= 0.78164 - 0.06858 = 0.71306. \end{aligned}$$

b) Adams-Moulton method:

$$\begin{aligned} y_4^p &= y_3 + \frac{h}{24} \left[55f_3 - 59f_2 + 37f_1 - 9f_0 \right] \\ &= y_3 + \frac{0.1}{24} \left[55(0.3 - 0.78164) - 59(0.2 - 0.83747) + 37(0.1 - 0.90968) + 9 \right] \\ &= 0.78164 - 0.04099 = 0.74065. \end{aligned}$$

$$\begin{aligned} y_4^c &= y_3 + \frac{h}{24} \left[9f_4 + 19f_3 - 5f_2 + f_1 \right] \\ &= y_3 + \frac{0.1}{24} \left[9(-0.34065) + 19(-0.48164) - 5(-0.63747) + (-0.80968) \right] \\ &= 0.78164 - 0.04100 = 0.74064. \end{aligned}$$

$$\begin{aligned}
 y_5^p &= y_4 + \frac{h}{24} [55f_4 - 59f_3 + 37f_2 - 9f_1] \\
 &= y_4 + \frac{0.1}{24} [55(-0.34064) - 59(-0.48164) + 37(-0.63747) + 9(0.80968)] \\
 &= 0.74064 - 0.02757 = 0.71307.
 \end{aligned}$$

$$\begin{aligned}
 y_5^c &= y_4 + \frac{h}{24} [9f_5 + 19f_4 - 5f_3 + f_2] \\
 &= y_4 + \frac{h}{24} [9(-0.21307) + 19(-0.34064) - 5(-0.48164) - 0.63747] \\
 &= 0.74064 - 0.02758 = 0.71306.
 \end{aligned}$$

The corresponding results given by analytical solution of the given differential equation correct to 5 decimal places are

$$y(x_4) = y(0.4) = 0.74064 \text{ and } y(x_5) = y(0.5) = 0.71306.$$

We note that values of Milne's and Adams-Moulton methods agree with the true values.

However, if we take $y_1 = 0.9095, y_2 = 0.8372, y_3 = 0.7813$, the values computed by using modified Euler's method, which are less accurate as compared to those obtained by using fourth-order Runge-Kutta method, as starting values for Milne's and Adams-Moulton methods we obtain y_4, y_5 which are tabulated below. These values are less accurate as compared to previous ones.

Table 12.3 Comparison of Results of Milne & Adams-Moulton Methods

n	x_n	$y(x_n)$ True value	Milne's method			Adams-Moulton method		
			y_n^p	y_n^c	$y(x_n) - y_n^c$ Error	y_n^p	y_n^c	Error
4	0.4	.74064	.74075	.74043	0.00021	.74035	.74034	.0003
5	0.5	.71306	.71297	.71277	0.00029	.71279	.71279	.00027

The above discussion shows that the accuracy of the results of predictor-corrector methods is based on the quality of starting values.

EXERCISE 12.5

1. Use Adams-Moulton method to find y_5 for the differential equation

$$\frac{dy}{dx} = x - 2y,$$

given the following data:

x		0.2	0.4	0.6
y	0.75000	0.52032	0.39933	0.35119

2. Solve the initial-value problem

$$\frac{dy}{dx} = y \cos x - 1, y(0) = 1, 0 \leq x \leq 0.5$$

using

- a) Milne's method b) Adams-Moulton method.

Working to 4 decimal places, evaluate starting values from the Euler's method with $h = 0.1$

3. Solve the initial-value problem

$$\frac{dy}{dx} = \cos x - \sin x + y, 0 \leq x \leq 0.5, y(0) = 0$$

using

- a) Milne's method b) Adams-Moulton method.

Obtain starting values by improved Euler's method with $h=0.1$.

4. solve the differential equation

$$\frac{dy}{dx} + y - e^{-x} \cos x = 0, 0 \leq x \leq 0.5, y(0) = 0$$

by

- a) Milne's method with $h = 0.1$.
b) Adams-Moulton method with $h = 0.1$.

12.7 CONVERGENCE AND STABILITY

All the numerical methods discussed so far are based on an equation that relates y_{n+1} to the preceding value y_n . Such an equation is known as a difference equation. When we apply a numerical method, we do not solve the given differential equation, but an equation related to it, i.e., the difference equation. A difference equation is not exactly equivalent to the

differential equation and consequently gives rise to the local truncation errors. In addition to truncation errors, the calculation of y_{n+1} will also be affected by rounding errors because, in general, it is impossible to work with exact arithmetic. Such errors incurred at one stage are carried forward automatically to subsequent stages. Even if a small step length is used to reduce truncation errors and high arithmetic precision is used to reduce rounding errors, it is possible for the errors to accumulate to such an extent to sweep out the solution. When this happens, the numerical method is said to be **unstable**.

12.7.1 CONVERGENCE AND STABILITY OF SINGLE-STEP METHODS

All the single step methods of sections 12.2-12.5 have an equation of the form

$$y_{n+1} = y_n + h \psi(x_n, y_n, h). \quad (12.24)$$

For example, in the improved Euler's method,

$$\begin{aligned} \psi(x_n, y_n, h) &= \frac{1}{2} \left[f(x_n, y_n) + f(x_{n+1}, y_{n+1}) \right] \\ &= \frac{1}{2} \left[f(x_n, y_n) + f(x_n + h, y_n + hf(x_n, y_n)) \right]. \end{aligned}$$

Equation (12.24) is known as a first order difference equation. The solution of this difference equation approximates the solution of the given differential equation. Rearranging (12.24) we obtain

$$\frac{y_{n+1} - y_n}{h} = \psi(x_n, y_n, h). \quad (12.25)$$

In fact when using the single-step method (12.24), the differential equation (12.2) is replaced by the difference equation (12.25). It is, therefore, important that (12.25) is a good approximation to (12.2). If the two equations were equivalent, the exact solution of the differential equation would also satisfy the difference equation. Thus, we define

$$T(x, h) = \frac{y(x+h) - y(x)}{h} - \psi(x, y, h) \quad (12.26)$$

as the local truncation error in the representation of the differential equation. For the numerical method to be useful, it is required that $T(x,h) \rightarrow 0$ as $h \rightarrow 0$.

DEFINITION 12.2: The single-step method using Equation (12.24) is said to be consistent with the differential equation (12.2) (i.e. the differential equation it approximates) if

$$\lim_{h \rightarrow 0} T(x,h) = 0, \quad \forall x \in [a,b].$$

Assuming that $\psi(x,y,h)$ is continuous then it follows from (12.2) and (12.6) that, for a consistent method

$$\psi(x,y,0) = f(x,y).$$

DEFINITION 12.3: The single step method of Equation (12.24) is said to be convergent with respect to the differential equation it approximates if

$$\lim_{h \rightarrow 0} \max_{1 \leq n \leq N} |y(x_n) - y_n| = 0,$$

where $y(x_n)$ denotes the exact solution of the differential equation and y_n the approximation obtained from the equation (12.24) at the n th step.

Let us discuss the stability of a single-step method. We will try to determine which methods are stable in the sense that small changes in the initial conditions produce correspondingly small changes in the subsequent approximations; that is, a stable method will be one which depends continuously upon the initial data.

THEOREM 12.2: If the initial-value problem

$$\frac{dy}{dx} = f(x,y), \quad y(a) = y_0$$

is approximated by a single-step method

$$y_{n+1} = y_n + h\psi(x_n, y_n, h)$$

and $\psi(x,y,h)$ satisfies a Lipschitz condition in the variable y on the set

$$D = \left\{ (x,y,h) \mid a \leq x \leq b, -\infty < y < \infty, 0 < h < h_0 \right\}$$

then the method is stable.

The outline of proof of this theorem is given in R. L. Burden [3]. The proof of the following theorem concerning the convergence of the single-step method can be found in Gear [5].

THEOREM 12.3 If the initial-value problem

$$\frac{dy}{dx} = f(x, y), \quad y(a) = y_0,$$

is approximated by a single-step method

$$y_{n+1} = y_n + h\psi(x_n, y_n, h)$$

and $\psi(x, y, h)$ is continuous and satisfies a Lipschitz condition in y on the set

$$D = \left\{ (x, y, h) \mid a \leq x \leq b, -\infty < y < \infty, 0 \leq h \leq h_0 \right\},$$

then the single-step method is convergent iff it is consistent.

The convergence property is important but it does not guarantee that a numerical method will produce an accurate solution to an initial-value problem. We analyze the improved Euler's method for convergence and stability.

EXAMPLE 12.7: Consider improved Euler's method given by

$$\begin{aligned} y_{n+1} &= y_n + \frac{h}{2} \left[f(x_n, y_n) + f(x_{n+1}, y_{n+1}) \right] \\ &= y_n + \frac{h}{2} \left[f(x_n, y_n) + f(x_n + h, y_n + hf(x_n, y_n)) \right], \quad n=0, 1, \dots, N-1. \end{aligned}$$

Here

$$\psi(x, y, h) = \frac{1}{2} [f(x, y) + f(x+h, y+hf(x, y))]$$

For $h = 0$,

$$\psi(x, y, 0) = \frac{1}{2} [f(x, y) + f(x, y)] = f(x, y),$$

and hence the consistency condition holds. Now

$$\begin{aligned} \psi(x, y, h) - \psi(x, \bar{y}, h) &= \frac{1}{2} [f(x, y) + f(x+h, y+hf(x, y))] - \\ &\quad \frac{1}{2} [f(x, \bar{y}) + f(x+h, \bar{y}+hf(x, \bar{y}))] \end{aligned}$$

$$\psi(x, y, h) - \psi(x, \bar{y}, h) = \frac{1}{2} [f(x, y) - f(x, \bar{y})] + \frac{1}{2} [f(x+h, y+hf(x, y)) - f(x+h, \bar{y}+hf(x, \bar{y}))]$$

or

$$|\psi(x, y, h) - \psi(x, \bar{y}, h)| \leq \frac{1}{2} |f(x, y) - f(x, \bar{y})| + \frac{1}{2} |f(x+h, y+hf(x, y)) - f(x+h, \bar{y}+hf(x, \bar{y}))|.$$

If f satisfies a Lipschitz condition on $\{(x, y) | a \leq x \leq b, -\infty < y < \infty\}$ with Lipschitz constant L , then

$$\begin{aligned} |\psi(x, y, h) - \psi(x, \bar{y}, h)| &\leq \frac{1}{2} L |y - \bar{y}| + \frac{1}{2} L |y + hf(x, y) - \bar{y} - hf(x, \bar{y})| \\ &\leq \frac{1}{2} L |y - \bar{y}| + \frac{1}{2} L |y - \bar{y} + h(f(x, y) - f(x, \bar{y}))| \\ &\leq \frac{1}{2} L |y - \bar{y}| + \frac{1}{2} L |y - \bar{y}| + \frac{1}{2} hL |f(x, y) - f(x, \bar{y})| \\ &\leq L |y - \bar{y}| + \frac{h}{2} L^2 |y - \bar{y}| \\ &\leq (L + \frac{1}{2} hL^2) |y - \bar{y}|. \end{aligned}$$

Therefore, ψ satisfies a Lipschitz condition in y on the set

$$\{(x, y, h) | a \leq x \leq b, -\infty < y < \infty, 0 \leq h \leq h_0\}$$

for any $h_0 > 0$ with Lipschitz constant $(L + \frac{1}{2} h_0 L^2)$.

Finally, if f is continuous on $\{(x, y) | a \leq x \leq b, -\infty < y < \infty\}$, then ψ is continuous on

$$\{(x, y, h) | a \leq x \leq b, -\infty < y < \infty, 0 \leq h \leq h_0\}.$$

Hence, Theorems 12.2 and 12.3 imply that the improved Euler's method is convergent and stable.

12.7.2 CONVERGENCE AND STABILITY OF MULTISTEP METHODS

The multistep methods use a difference equation which is a good approximation to the differential equation. The problems involved with consistency, convergence, and stability of multistep methods are compounded because of the number of approximations involved at each step. In the single-step methods, the approximation y_{n+1} depends only on the previous approximation y_n , while the multistep methods use at least two of the previous approximations.

The concept of convergence for multistep methods is the same as that for single-step methods; i.e., a multistep method is convergent if the solution of the difference equation tends to the solution of the differential equation as the step size tends to zero. Symbolically it can be written as

$$\lim_{h \rightarrow 0} \max_{1 \leq n \leq N} |y(x_n) - y_n| = 0.$$

Stability in a multistep method means that, for sufficiently small step sizes, small changes in any or all starting values produce only small changes in the subsequent values.

The multistep methods use a difference equation of the form

$$\sum_{j=0}^l a_j y_{m+j} = h \sum_{j=0}^l b_j y_{m+j} \quad (12.27)$$

where a_j and b_j are constants and l is the number of steps.

When $b_l = 0$, the method is called an explicit or open method; otherwise the method is called an implicit or closed method. For example, the Adams-Moulton predictor formula (12.22') can be written as

$$-y_n + y_{n+1} = \frac{h}{24} \left[55f_n - 59f_{n-1} + 37f_{n-2} - 9f_{n-3} \right]$$

and so, in this case,

$$\begin{aligned} l &= 4, m = n - 3 \\ a_0 &= 0, a_1 = 0, a_2 = 0, a_3 = -1, a_4 = 1 \\ b_0 &= -9/24, b_1 = 37/24, b_2 = -59/24, b_3 = 55/24, b_4 = 0. \end{aligned}$$

Since $b_4 = 0$, the method is explicit.

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and so, in this case,

$$l = 4, m = n - 3 \\ a_0 = 0, a_1 = 0, a_2 = 0, a_3 = -1, a_4 = 1 \\ b_0 = -9/24, b_1 = 37/24, b_2 = -59/24, b_3 = 55/24, b_4 = 0.$$

Since $b_4 = 0$, the method is explicit.

Likewise, the Adams-Moulton corrector formula (12.23') could be expressed in the form of (12.27) by choosing $l=3$, $m=n-2$, $a_0=0$, $a_1=0$, $a_2=-1$, $a_3=1$, $b_0=1/24$, $b_1=-5/24$, $b_2=19/24$, $b_3=9/24$.

This method is implicit as $b_3 \neq 0$.

The local truncation error for the multistep methods using Eq. (12.27) is

$$\begin{aligned} T_k &= \sum_{j=0}^l a_j y_{m+j} - h \sum_{j=0}^l b_j f_{m+j} \\ &= \sum_{j=0}^l a_j y(x + jh) - \sum_{j=0}^l b_j f(x + jh). \end{aligned}$$

A polynomial associated with the difference equation (12.27) is

$$P(\lambda) = \sum_{j=0}^l a_j \lambda^j$$

which is called the characteristic polynomial of the method.

DEFINITION 12.4: Let $\lambda_1, \lambda_2, \dots, \lambda_l$ be the (not necessarily distinct) roots of the characteristic polynomial equation

$$P(\lambda) = 0$$

associated with the multistep method using Eq. (12.27). If $|\lambda_i| \leq 1$ for each $i = 1, 2, \dots, l$, and the roots with absolute value 1 are simple, then the multistep method is said to be zero-stable or to satisfy the root condition.

The concepts of consistency, root condition, convergence and stability of a method are drawn together in the following theorem. For the proof of this result, the reader is referred to Isaacson and Keller [7].

THEOREM 12.4: A multistep method using the difference equation (12.27) is stable iff it satisfies the root condition. Moreover, if the multistep method is consistent with the differential equation, then the method is stable iff it is convergent.

The methods which satisfy the root condition and have only one root with magnitude one are called **strongly stable**, while those having more than one root with magnitude one are called **weakly stable**.

EXAMPLE 12.8: We have seen that for the Adams-Moulton predictor formula (12.22'), $l=4$. Thus the characteristic polynomial for the Adams-Moulton predictor formula is

$$P(\lambda) = \sum_{j=0}^4 a_j \lambda^j = a_0 + a_1 \lambda + a_2 \lambda^2 + a_3 \lambda^3 + a_4 \lambda^4.$$

But, in this case $a_0 = 0$, $a_1 = 0$, $a_2 = 0$, $a_3 = -1$, and $a_4 = 1$.

Therefore, $P(\lambda) = \lambda^4 - \lambda^3 = \lambda^3(\lambda - 1)$,

and the root of the equation $P(\lambda) = 0$ are $\lambda_1 = 0$, $\lambda_2 = 0$, $\lambda_3 = 0$, and $\lambda_4 = 1$.

Thus the Adams-Moulton predictor formula satisfies the root condition and is stable. Since there is only one root with magnitude one i.e. $\lambda_4 = 1$, the method is strongly stable.

The Adams-Moulton corrector formula has a characteristic polynomial equation $P(\lambda) = \lambda^3 - \lambda^2 = 0$, which has roots $\lambda_1 = 0$, $\lambda_2 = 0$, $\lambda_3 = 1$, and satisfies the root condition and hence is stable. Again, as there is only one root with magnitude one, the method is strongly stable.

The Milne's corrector formula (12.21') can be expressed as

$$-y_{n-1} + y_{n+1} = \frac{h}{3} [f_{n+1} + 4f_n + f_{n-1}],$$

so $l = 2$, $m = n - 1$,

and $a_0 = -1$, $a_1 = 0$, $a_2 = 1$, $b_0 = 1/3$, $b_1 = 4/3$, $b_2 = 1/3$.

the characteristic polynomial equation is

$$P(\lambda) = 0$$

or

$$a_0 + a_1 \lambda + a_2 \lambda^2 = 0$$

or

$$\lambda^2 - 1 = 0 \Rightarrow \lambda_1 = 1, \lambda_2 = -1.$$

Thus the Milne's corrector formula also satisfies the root condition and is stable. As its characteristic polynomial equation has two roots with magnitude one, the method is weakly stable.